

Queue Cure – Case Study

Executive Summary

Queue Cure is an AI-powered real-time clinic queue management system designed to eliminate paper tokens, reduce patient uncertainty, and improve clinic operations through live queue tracking, QR-based waiting rooms, and AI-assisted triage.

Problem Statement

Neighborhood clinics often rely on paper tokens and manual patient calling. Patients have little visibility into queue progress and reception staff repeatedly answer questions regarding wait times and queue position. Urgent cases may not receive timely attention.

Solution

Queue Cure digitizes the entire queue workflow. Receptionists register patients, AI classifies complaints into ROUTINE, URGENT, or EMERGENCY categories, QR codes provide instant access to a patient waiting room, and real-time synchronization keeps every screen updated.

Key Features

- Real-time queue synchronization via Supabase Realtime CDC
- AI-assisted triage using Groq LLaMA
- QR-code based patient waiting rooms
- Smart wait-time estimation
- Queue state machine with audit trail
- Multi-clinic architecture

Technical Architecture

Frontend: Next.js 14 Backend: Supabase PostgreSQL Realtime Layer: WebSockets via Supabase Realtime CDC AI Layer: Groq LLaMA Deployment: Vercel + Supabase Edge Functions

Development Process

The project began with workflow mapping and pain-point analysis. A polling-based design was initially considered but replaced with an event-driven architecture for lower latency and better scalability. State-machine design ensured predictable queue behavior and robust edge-case handling.

Results & Outcomes

- Sub-200ms synchronization across connected devices
- Zero page refreshes required for queue updates
- Real-time queue visibility for patients
- Reduced receptionist workload through self-service tracking
- AI-supported prioritization of urgent cases

Challenges & Learnings

Medical triage requires careful validation and should assist—not replace—clinical judgment. Building reliable realtime systems requires strong state management, conflict prevention, and resilience handling.

Future Improvements

Doctor dashboard, appointment scheduling, advanced analytics, predictive wait-time forecasting, offline resilience, and large-scale multi-clinic deployment support.

Conclusion

Queue Cure demonstrates how AI and realtime technologies can modernize healthcare operations, improve patient experience, and create a scalable foundation for digital clinic management.